

Functional Data Analysis of Tecator Spectra

FPCA and Depth-Based Outlier Detection



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Functional and Topological Data Analysis



Project Aim

The aim is to analyze Tecator spectra as functional data and detect atypical curves.

- 01 Can spectra be treated as functional observations?
- 02 What are the main modes of variation?
- 03 Can depth measures detect atypical curves?
- 04 Are outliers related to fat, water, or protein?



Tecator Dataset

215
spectra

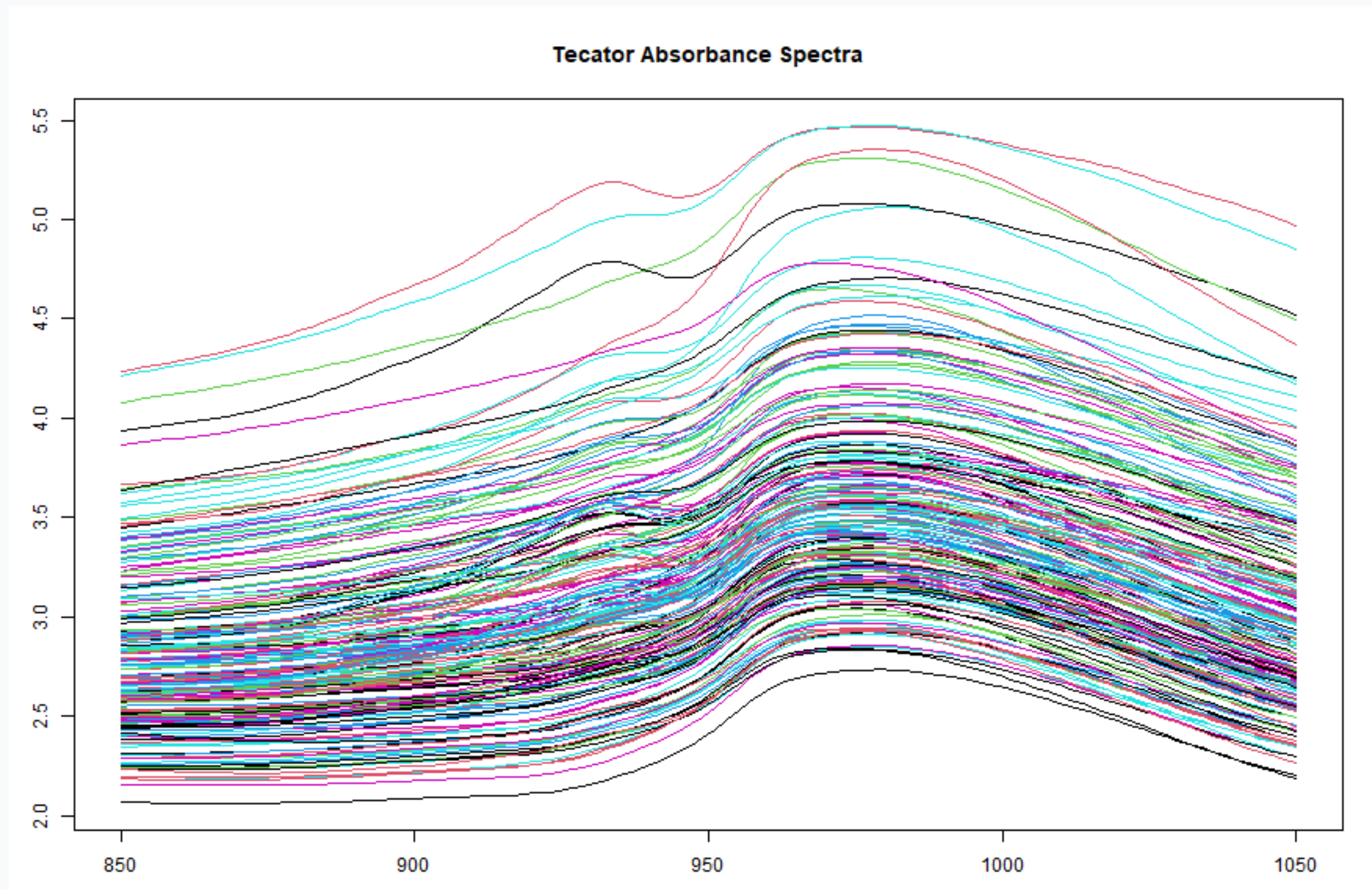
100
wavelength points

3
chemical variables

Variables	fat, water, protein
Domain	wavelength
Observation	absorbance curve

Each meat sample is represented by an entire curve $x_i(t)$

Why This Is Functional Data?



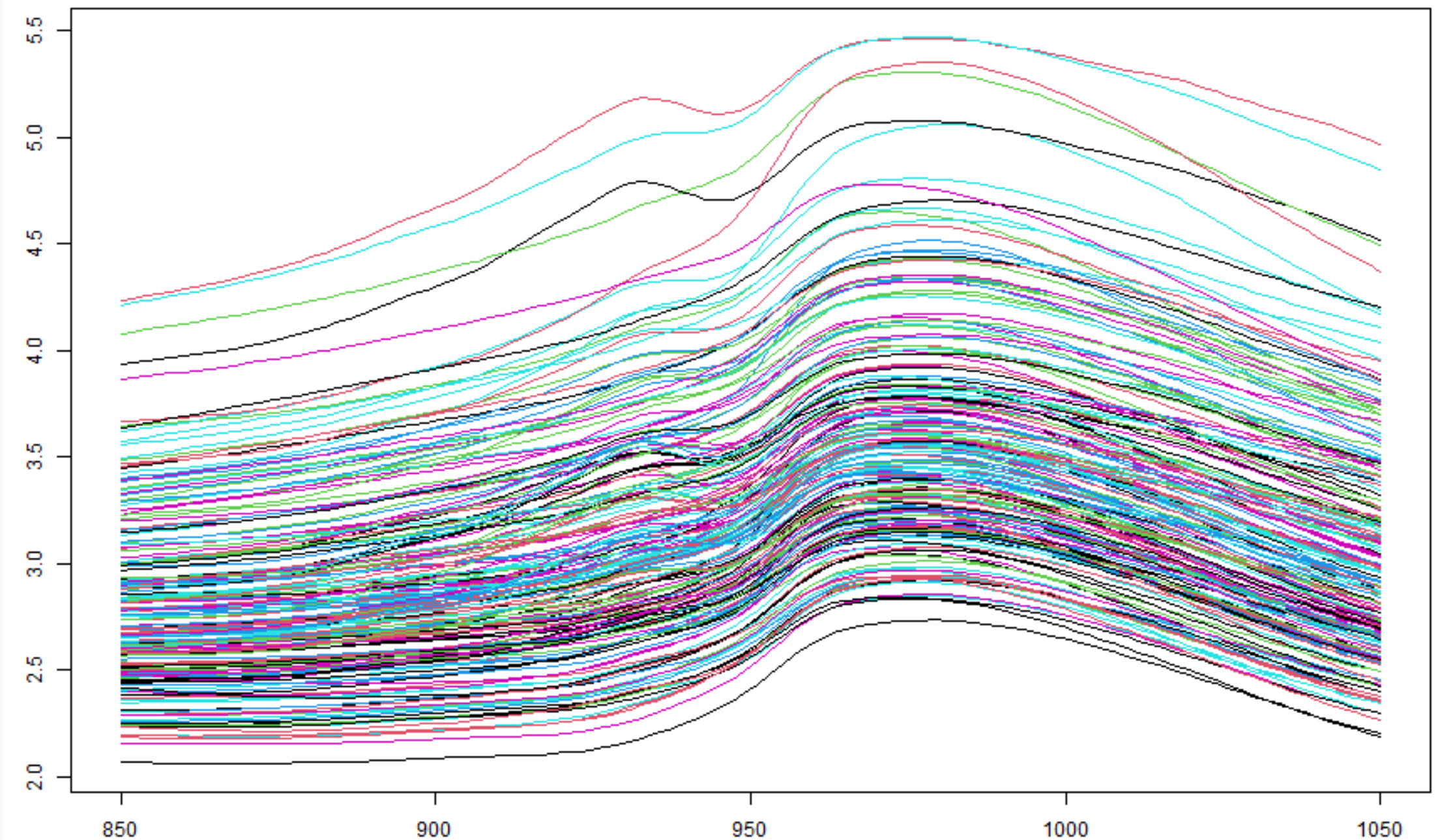
Each sample is not
a single number, but
a curve over
wavelength

$$\mathbf{x}_i(\mathbf{t}), \mathbf{t} \in \mathbf{T}$$

Functional Representation with B-Splines

Raw spectra are observed at discrete wavelength points.
FDA represents them as smooth functions

$$x_i(t) \approx \sum c_{ik} \varphi_k(t)$$

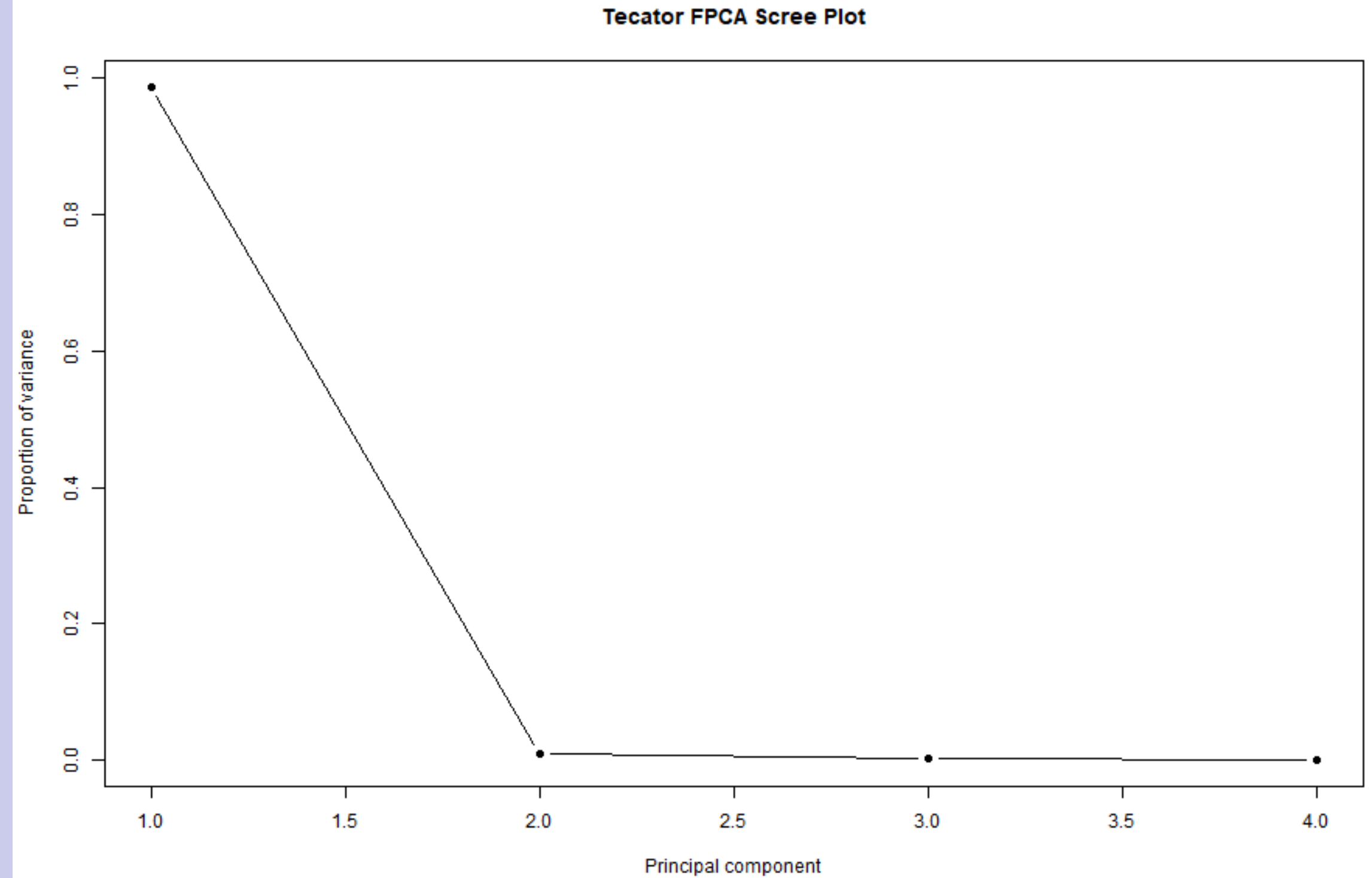


B-spline basis: 35 basis functions, cubic splines.

Registration was not needed because all spectra share the same wavelength grid

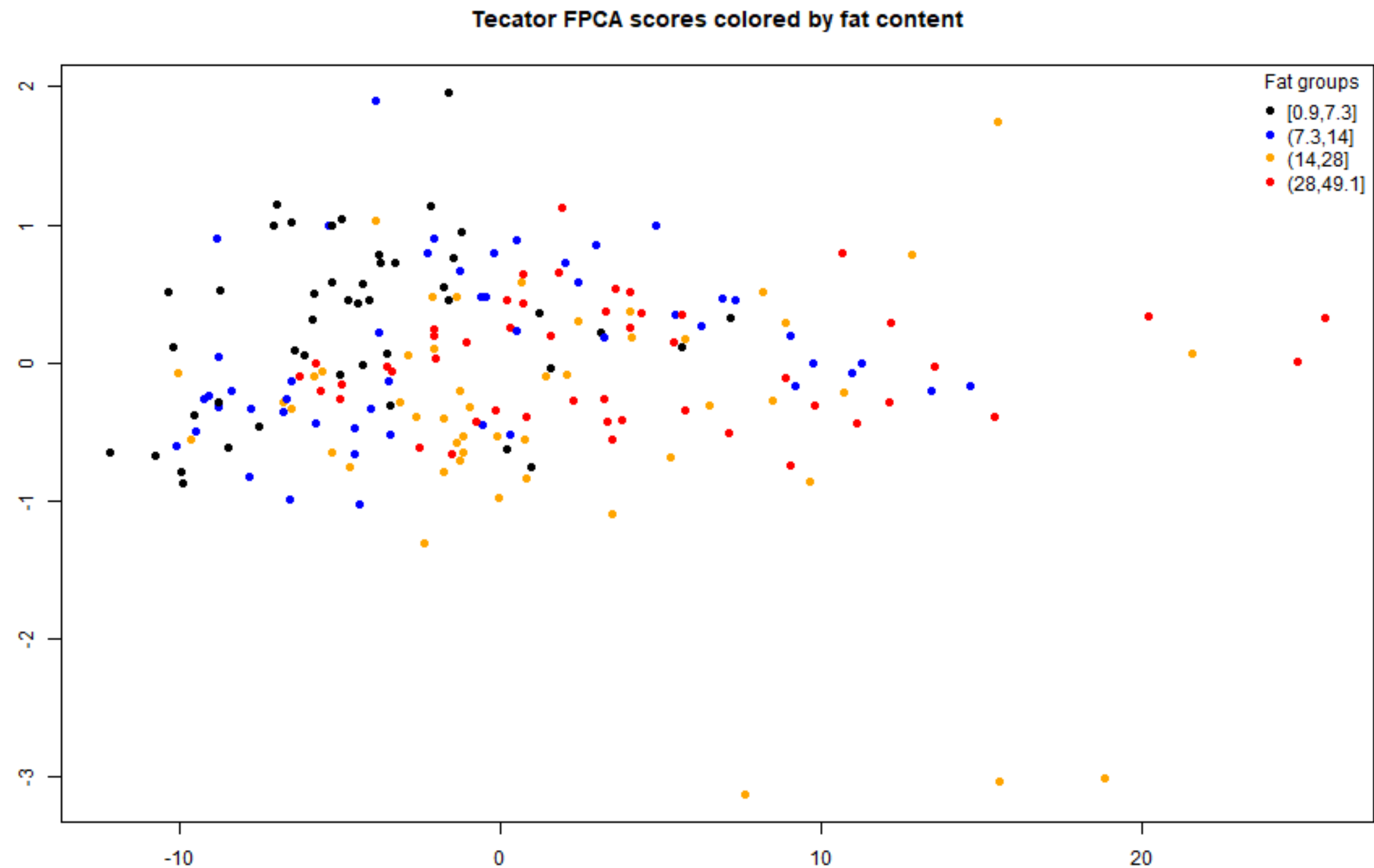
PC1 explains almost all functional variability

FPCA: Main Functional Variation



Later components are secondary and should be interpreted cautiously

FPCA Scores and Chemical Composition



	Fat	Water	Protein
PC1	0.44	-0.48	-0.26
PC2	-0.14	0.13	0.36
PC3	0.62	-0.65	-0.47
PC4	-0.53	0.47	0.52

FPCA scores are associated with chemical composition, but this is not causal

Depth Measures: From Central to Outlying Curves

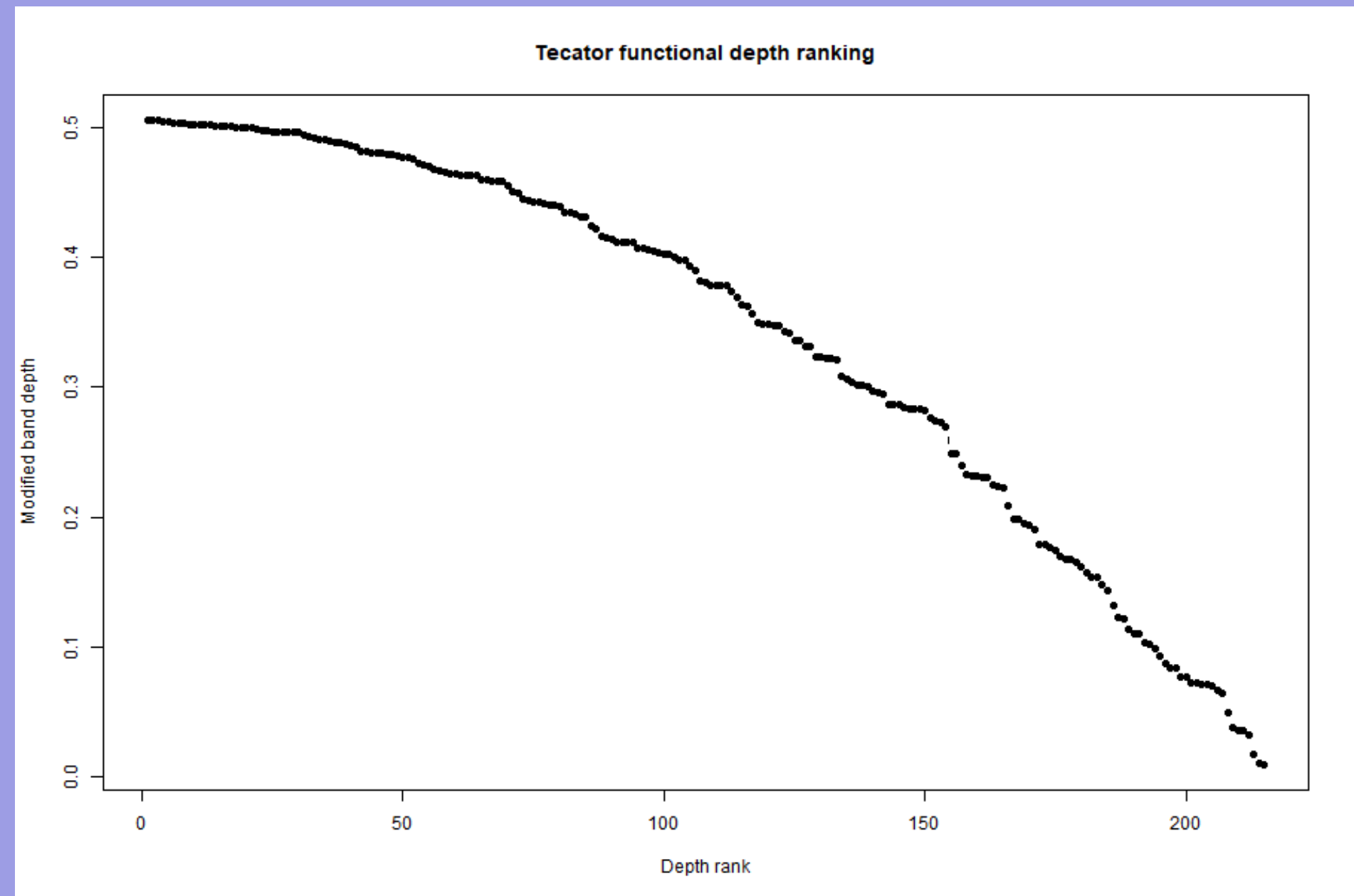
Curves cannot be naturally ordered like scalar data.

Depth assigns each curve a centrality value.

High depth $\rightarrow$ central curve.

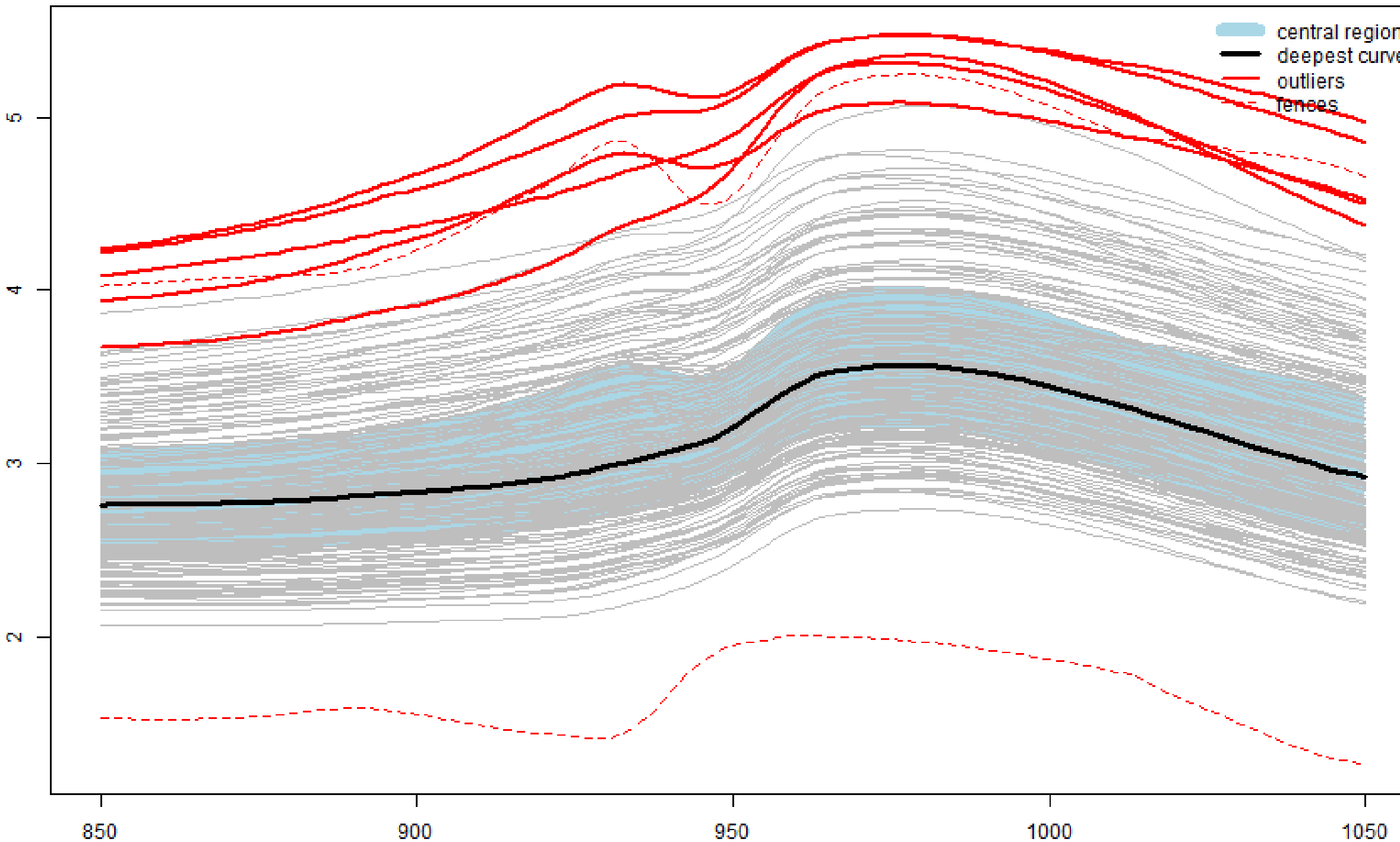
Low depth $\rightarrow$ possible outlier.

Modified Band Depth ranks curves by how often they lie inside bands formed by other curves



Depth-Based Outlier Detection

Tecator Functional Boxplot



The functional boxplot identifies spectra whose level or shape differs from the central pattern

Detected outliers

43

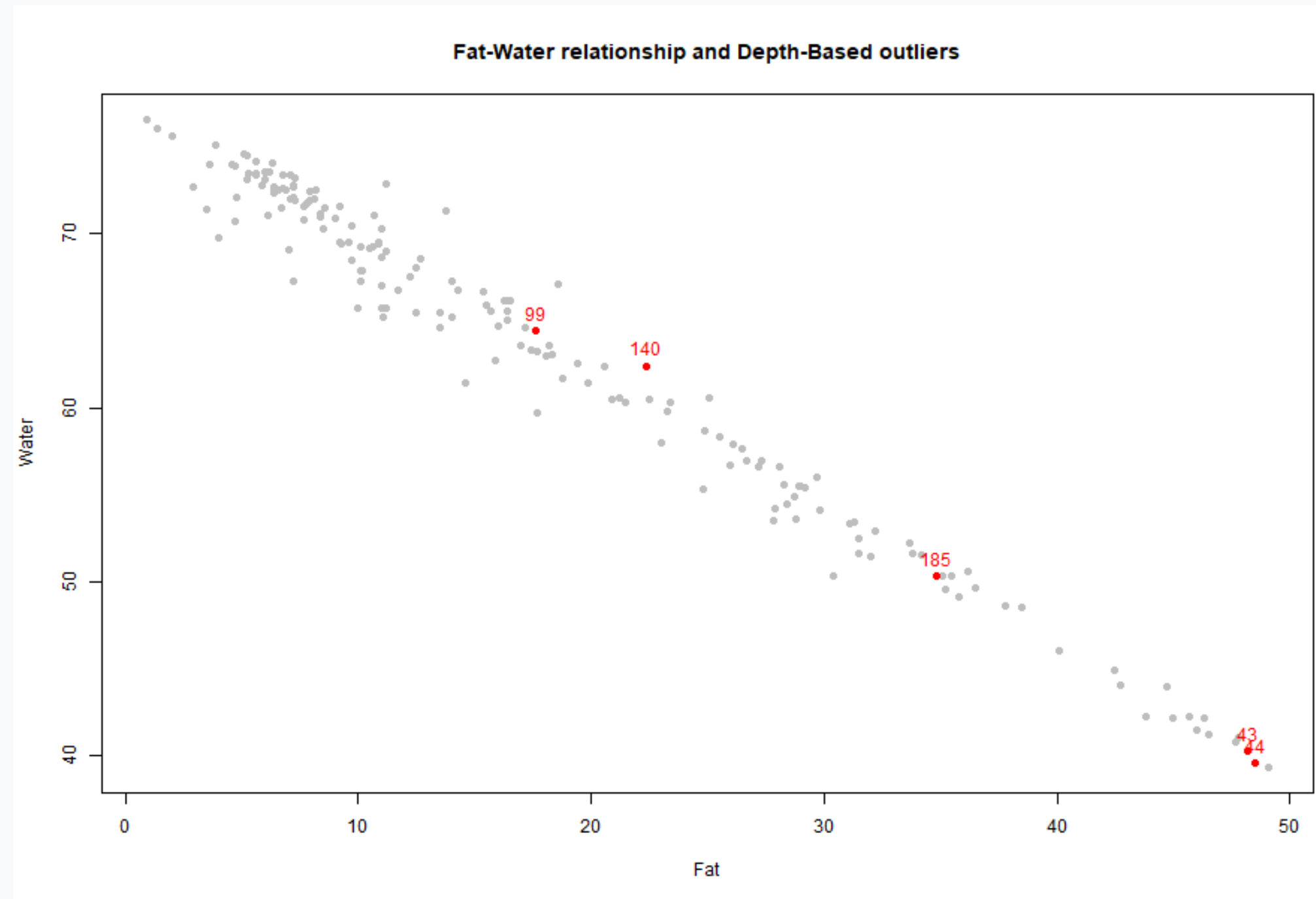
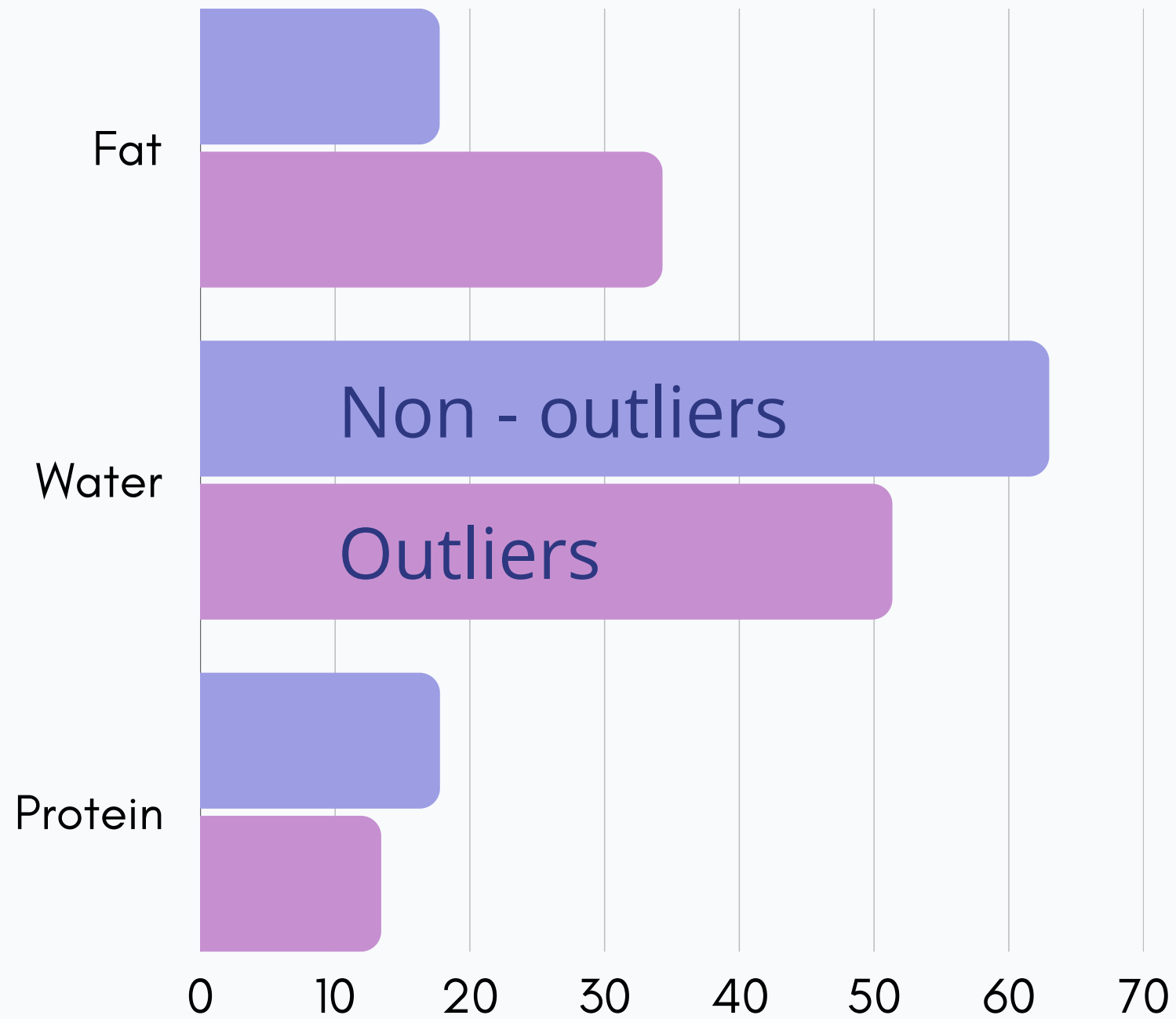
44

99

140

185

Outliers Are Chemically Meaningful



Depth-based outliers have higher average fat and lower average water/protein

FPCA and depth measures provide complementary tools for exploratory FDA

Thank you for your attention

FPCA and Depth-Based Outlier Detection

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